

Lierda L-LRMAM36-FANN4 Module Hardware Design Manual

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Document Revision History

Document version	Change Date	Reviser	Reviewer	Change content
Rev1.0	26-06-02	CJH&JHB	LXQ	Initial version

Safety Instructions

Users are responsible for complying with the regulations of other countries regarding wireless communication modules and devices, as well as the regulations applicable to specific usage environments. By following the safety principles below, personal safety can be ensured, and products and working environments can be protected from potential damage. Our company assumes no responsibility for any related losses caused by customers' failure to comply with these regulations.



Road safety comes first! When driving, do not use handheld mobile terminal devices unless they have a hands-free function. Please park before making a call!



Please turn off mobile terminal devices before boarding. The wireless functions of mobile terminals are prohibited from being turned on during the flight to prevent interference with the aircraft's communication systems. Ignoring this warning may endanger flight safety and may even violate the law.



When in hospitals or healthcare facilities, pay attention to any restrictions on the use of mobile terminal devices. RF interference may cause medical equipment to malfunction, so it may be necessary to turn off mobile terminal devices.



Mobile terminal devices do not guarantee that an effective connection can be made under all circumstances, such as when the mobile terminal device has no phone credit or the SIM card is invalid. If you encounter the above situations in an emergency, please remember to use emergency calling, and ensure that your device is powered on and in an area with sufficient signal strength.



Your mobile terminal device receives and transmits radio frequency signals when powered on, and may cause radio frequency interference when near televisions, radios, computers, or other electronic devices.



Please keep the mobile terminal device away from flammable gases. When you are near gas stations, oil depots, chemical plants, or blasting operation sites, please turn off the mobile terminal device. Operating electronic devices in any potentially explosive environment poses safety risks.

Applicable Module Selection

Serial Number	Module model	Supported frequency bands	Dimensions (mm)	Module Introduction
1	L-LRMAM36-FANN4	863~930MHz(LoRa) 2400~2500MHz(WiFi&BLE)	20*20*2.5	No firmware / No antenna

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1 Introduction

This document mainly provides a detailed description of the hardware parameters, pin function descriptions, interface function definitions, mechanical dimensions, production process requirements, and other aspects of the L-LRMAM36-FANN4 module. By reading this document, users can quickly understand the hardware design-related content of the product.



Figure 1.1 Module Schematic Diagram

2 Product Overview

The Lierda L-LRMAM36-FANN4 module is a WiFi & BLE + LoRa module developed based on Espressif' s ESP32-S3 and Semtech' s fourth-generation LoRa (Generation 4 LoRa) RF integrated chip LR2021.

The module supports 2.4G Wi-Fi and BLE wireless communication. It integrates a high-performance Xtensa® 32-bit LX7 dual-core processor, as well as an ultra-low-power processor.

The module supports the latest Generation 4 LoRa, with a maximum power of up to 21 dBm in the Sub-G frequency band.

The module offers compact form factor, high throughput, low power, extended range, and robust interference immunity. It is highly suitable for applications requiring high transmission rates and long transmission distances, such as smart homes, security monitoring, smart communities, logistics and warehousing, industrial control, and other application scenarios.

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2.1 Product Features

Table 2-1 Key Features of the Module

Parameters	Description
Package Type	LCC/LGA
Wireless Technology	● IEEE 802.11b/g/n ● BLE 5.0 ● LoRa ● FLRC ● 2-FSK & 4-FSK
Wi-Fi bandwidth	Supports standard 20/40 MHz bandwidth
Module dimensions	20mm × 20mm × 2.5mm
Operating voltage	3.0 V to 3.6 V, typical value 3.3 V
Operating frequency band	2400 ~ 2483.5MHz (WiFi、 BLE) 863 ~ 930MHz (LoRa)
Antenna	No antenna
Operating temperature 1	-40 ~ +65℃
Storage temperature	-40 ~ +85℃
Internal storage	8MB Flash + 8MB PSRAM

Note

1. The operating ambient temperature of the module's main chip, ESP32-S3R8, is -40℃ to +65℃. If the PSRAM ECC function is enabled, the maximum ambient temperature of the module can be increased to 85℃, but the usable capacity of the PSRAM will be reduced by 1/16.

2.2 Functional block diagram

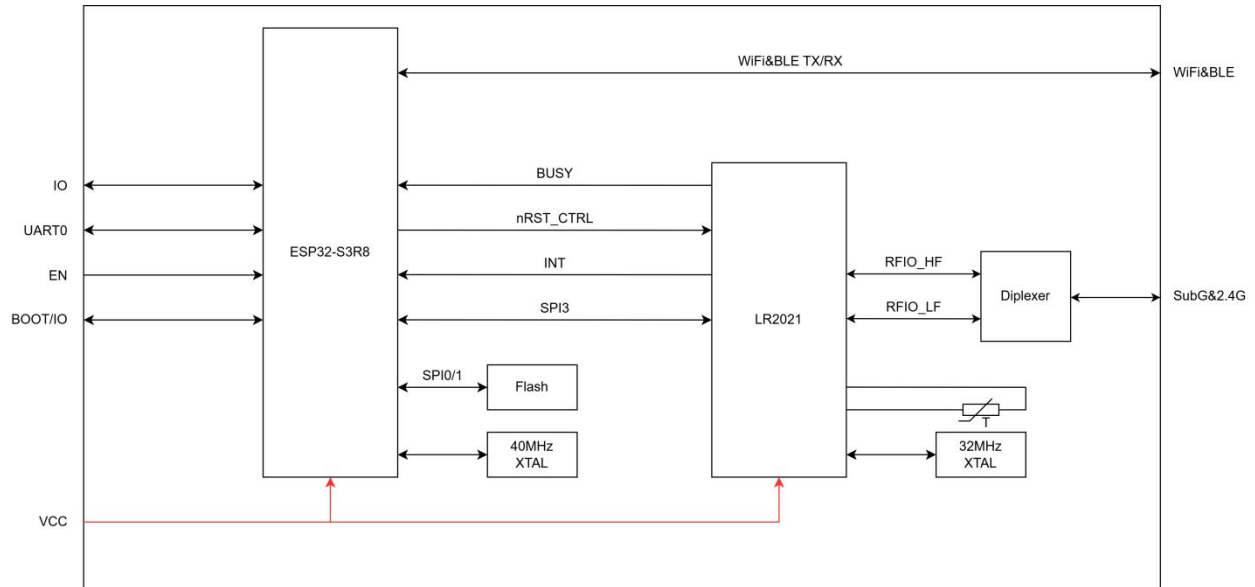


Figure 2.1 System Block Diagram

2.3 Pin Definition

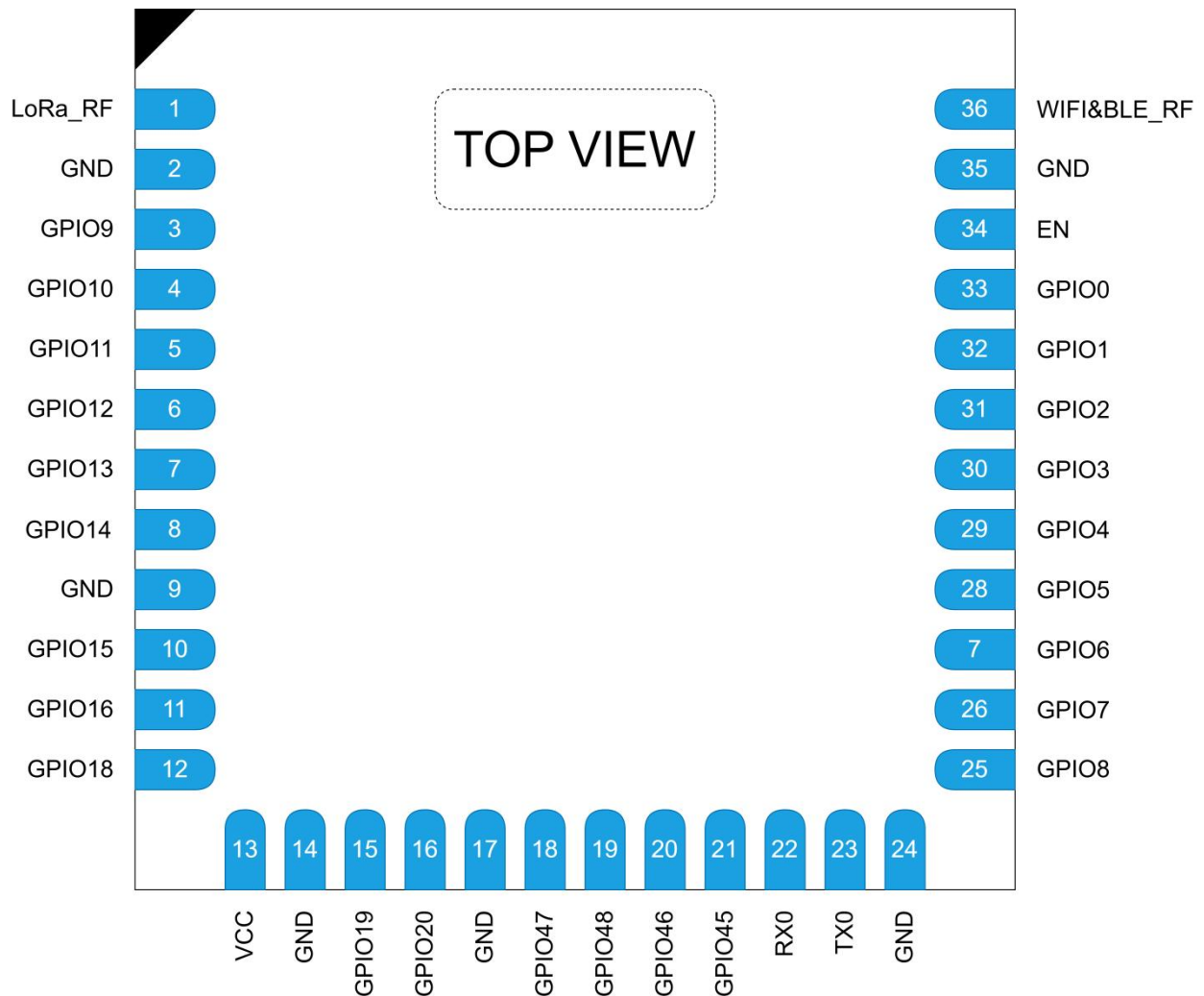


Figure 2.2 Module Pin Diagram

2.4 Pin Description Table

Table 2-2 Pin Function Description

Pin	Pin Definition	I/O Type	Voltage	Function Description
1	LoRa_RF	RF	-	LoRa RF
2	GND	G	-	GND
3	GPIO9	IO	VCC	RTC_GPIO9/GPIO9/TOUCH9/ADC1_CH8 /FSPIHD/SUBSPIHD
4	GPIO10	IO	VCC	RTC_GPIO10/GPIO10/TOUCH10/ADC1_CH9/FSPICS0/FSPIIO4,SUBSPICS0

5	GPIO11	IO	VCC	RTC_GPIO11/GPIO11/TOUCH11/ADC2_CH0/FSPID/FSPIIO5/SUBSPID
6	GPIO12	IO	VCC	RTC_GPIO12/GPIO12/TOUCH12/ADC2_CH1/FSPICLK/FSPIIO6,SUBSPICLK
7	GPIO13	IO	VCC	RTC_GPIO13/GPIO13/TOUCH13/ADC2_CH2/FSPIQ/FSPIIO7/SUBSPIQ
8	GPIO14	IO	VCC	RTC_GPIO14/GPIO14/TOUCH14/ADC2_CH3/FSPIWP/FSPIDQS,SUBSPIWP
9	GND	G	-	GND
10	GPIO15	IO	VCC	RTC_GPIO15/GPIO15/U0RTS/ADC2_CH4/XTAL_32K_P
11	GPIO16	IO	VCC	RTC_GPIO16/GPIO16/U0CTS/ADC2_CH5/XTAL_32K_N
12	GPIO18	IO	VCC	RTC_GPIO18/GPIO18/U1RXD/ADC2_CH7/CLK_OUT3
13	VCC	P	3.0~3.6V	Power input
14	GND	G	-	GND
15	GPIO19	IO	VCC	RTC_GPIO19/GPIO19/U1RTS/ADC2_CH8/CLK_OUT2/USB_D-
16	GPIO20	IO	VCC	RTC_GPIO20/GPIO20/U1CTS/ADC2_CH9/CLK_OUT1/USB_D+
17	GND	G	-	GND
18	GPIO47	IO	VCC	SPICLK_P_DIFF/GPIO47/SUBSPICLK_P_DIFF
19	GPIO48	IO	VCC	SPICLK_N_DIFF/GPIO48/SUBSPICLK_N_DIFF
20	GPIO46	IO	VCC	GPIO46
21	GPIO45	IO	VCC	GPIO45
22	RX0	IO	VCC	U0RXD/GPIO44/CLK_OUT2
23	TX0	IO	VCC	U0TXD/GPIO43/CLK_OUT1
24	GND	G	-	GND
25	GPIO8	IO	VCC	RTC_GPIO8/GPIO8/TOUCH8/ADC1_CH7/SUBSPICS1
26	GPIO7	IO	VCC	RTC_GPIO7/GPIO7/TOUCH7/ADC1_CH6
27	GPIO6	IO	VCC	RTC_GPIO6/GPIO6/TOUCH6/ADC1_CH5

28	GPIO5	IO	VCC	RTC_GPIO5/GPIO5/TOUCH5/ADC1_CH4
29	GPIO4	IO	VCC	RTC_GPIO4/GPIO4/TOUCH4/ADC1_CH3
30	GPIO3	IO	VCC	RTC_GPIO3/GPIO3/TOUCH3/ADC1_CH2
31	GPIO2	IO	VCC	RTC_GPIO2/GPIO2/TOUCH2/ADC1_CH1
32	GPIO1	IO	VCC	RTC_GPIO1/GPIO1/TOUCH1/ADC1_CH0
33	GPIO0	IO	VCC	RTC_GPIO0/GPIO0
34	EN	I	VCC	High level: chip enabled; Low level: Chip is disabled. Be careful not to leave the EN pin floating.
35	GND	G	-	GND
36	WIFI&BLE_RF	RF	-	WiFi & BLE RF

“P”:POWER “IO”:INPUT or OUTPUT “G”:GND

3 Operating Characteristics

3.1 Power

3.1.1 Power interface

This module provides 1 VCC pin for connecting to an external power supply. The interface description is shown in the table below:

Table 3-1 Power Pin Definitions

Pin number	Pin Definition	Description	Minimum value (V)	Typical value (V)	Maximum value (V)
13	VCC	Module power supply	3.0	3.3	3.6

3.1.2 Power Circuit Reference Design

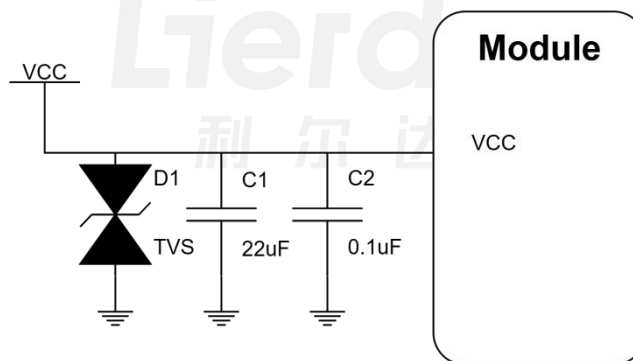


Figure 3.1 Module Power Supply Reference Circuit

VCC is the power supply input for the entire module. Power supply quality directly affects module RF performance. When the power supply ripple noise is too large, the WiFi EVM metric will deteriorate. During design, a power supply capable of providing at least 1A@3.3V must be selected to ensure that the module's supply input voltage will not fall below the minimum operating voltage under any circumstances, preventing abnormal module operation caused by voltage drops.

If the voltage difference between the input voltage and the module's supply voltage is

not very large, it is recommended to use an LDO as the power supply. If there is a relatively large voltage difference between the input and output, use a DC-DC converter for power conversion, while also paying attention to the EMI issues caused by the DC/DC converter.

In PCB design, the longer the VCC trace, the wider the trace should be. The trace must meet a current-carrying capacity of more than 1 A, and a trace width of 0.8 mm or more is recommended (with 1 oz copper thickness). The GND plane in the power supply section should be as complete as possible, with plenty of ground vias.

It is recommended to reserve two or more capacitors at the module's VCC power pin. The actual capacitance values can be adjusted according to the actual situation, and the capacitors should be placed as close as possible to the module's VCC pin.

Remarks

- D1 is used for electrostatic protection. Users need to pay attention to the product's ESD requirements during design; see Table 5-1 Product Absolute Maximum Ratings. If higher protection requirements are needed, it is recommended to place an external TVS diode for electrostatic protection.

3.2 Enable

3.2.1 Enable interface

Table 3-2 Enable Pin Definitions

Pin number	Pin Definition	Function Description
34	EN	High level: chip enabled. Low level: Chip is disabled. Do not leave the EN pin floating.

3.2.2 Enable Interface Reference Design

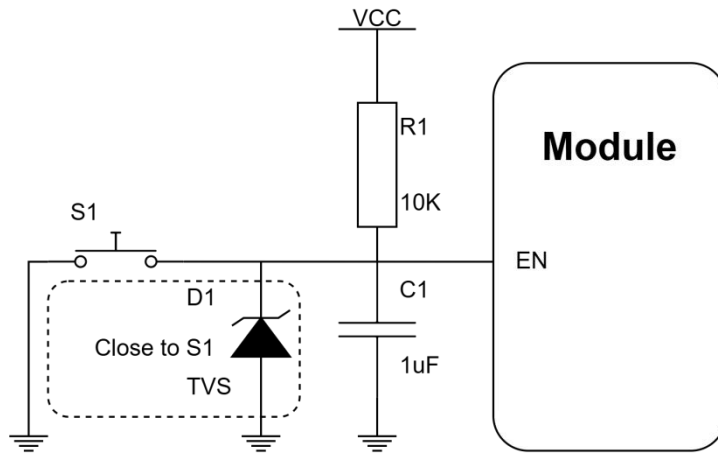


Figure 3.2 Reset Reference Circuit

After the main chip inside the module is powered on, its power rail requires some time to stabilize. After that, the EN pin used for power-on and reset is pulled high to activate the chip. To ensure normal power supply when the module is powered on, an RC delay circuit needs to be added at the EN pin. The recommended RC values are typically $R = 10\text{ k}\Omega$ and $C = 1\text{ }\mu\text{F}$, but the specific values still need to be adjusted according to the power-on timing of the module power supply and the power-on reset timing of the chip.

For more information about EN, power-on, and reset timing, see the figure and table below.

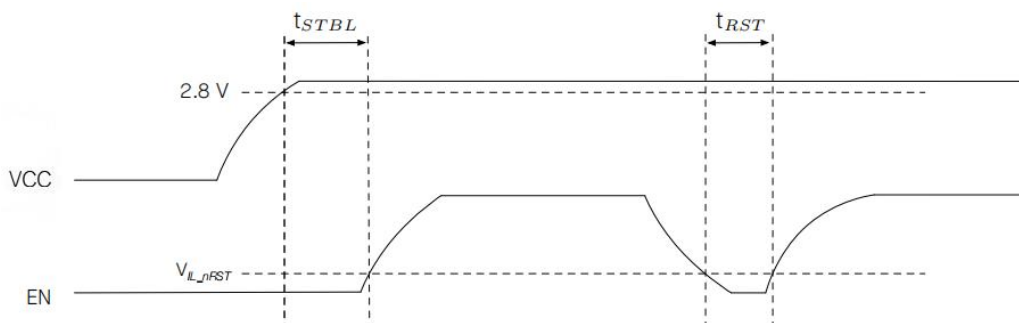


Figure 3.3 Reference Parameter Diagram for Power-On and Reset Timing

Table 3-4 Description of Power-On and Reset Timing Parameters

Instructions	Description	Minimum value (μs)
--------------	-------------	------------------------------------

t_{STBL}	The time required for VCC to stabilize before pulling the EN pin high to activate the chip.	50
t_{RST}	The time when the EN level is lower than V_{IL_nRST} (refer to Table 5-2 for the specific value) to reset the chip.	50

Remarks

- The reset line should not be routed too long. Pay attention to ground shielding protection, and keep it away from RF, VCC power supplies, and strong signal interference sources.

3.3 SPI

The ESP32-S3 chip used by the module has the following SPI interfaces:

- SPI0, used by GDMA and cache to access flash/PSRAM in-package or off-package flash/PSRAM.
- SPI1, for the CPU to access Flash/PSRAM inside or outside the package;
- SPI2, a general-purpose SPI controller, supports DMA access via GDMA channel allocation.
- SPI3, a general-purpose SPI controller, is accessed by allocating DMA channels through GDMA.

This module design recommends that users configure the SPI3 interface as the SPI interface used for communication between the ESP32-S3 and the LR2021.

For communication with other SPI peripherals, users can use the more feature-rich SPI2 interface to communicate with the peripherals.

3.3.1 Description of Internal Communication Between ESP32-S3 and LR2021

The internal main chip ESP32-S3 of the module communicates with the LR2021, which occupies some IO pins. Users need to configure and use these IO pins according to their

corresponding functions. The unused IO pins of the ESP32-S3 can be used normally by users.

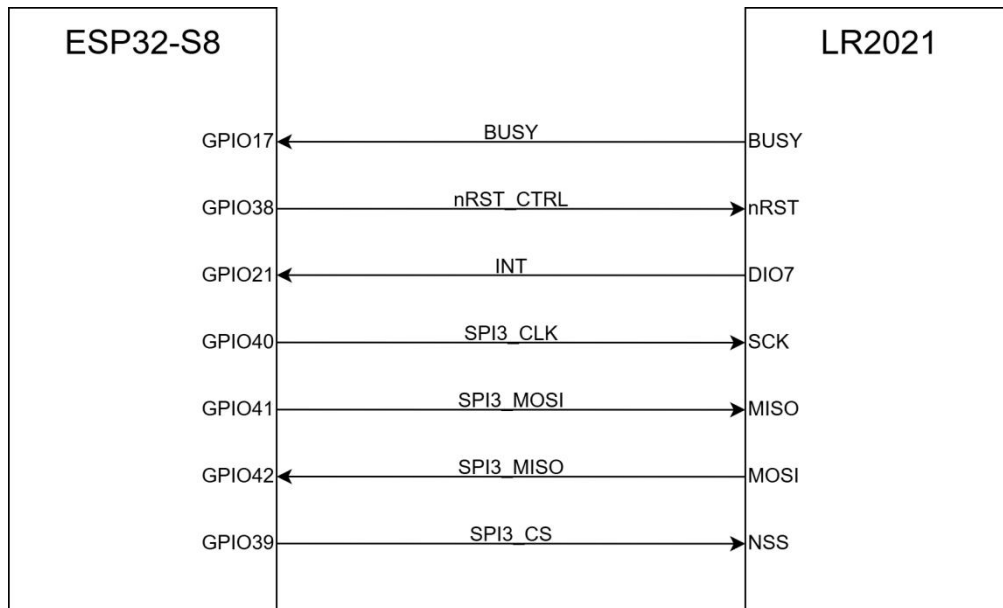


Figure 3.4 Internal Communication Circuit

Table 3-5 Internal Communication Function Description

ESP32-S3 GPIO	LR2021 Pins	Function Description
GPIO17	BUSY	Busy Indicator
GPIO38	nRST	LR2021 reset pin, active low.
GPIO21	DIO7	Interrupt source mapping pin
GPIO40	SCK	SPI clock input
GPIO41	MISO	SPI master data input, slave data output
GPIO42	MOSI	SPI master data output, slave data input
GPIO39	NSS	SPI chip select

3.3.2 SPI2 interface

It is recommended to use GPIO9 – GPIO14 as SPI2 pins for the module. First, they can be directly multiplexed through IO MUX, avoiding performance degradation caused by transmission through the GPIO matrix. Second, the module hardware design has also applied length matching to these traces for high-speed transmission.

Of course, when it is inconvenient to use GPIO9 to GPIO13, users can configure them as any GPIO through the GPIO swap matrix.

Table 3-6 Recommended Pin Configuration for SPI2

Pin number	Pin Definition	Recommended configuration	Feature Description
3	GPIO9	FSPIHD	Pause (Hold)
4	GPIO10	FSPICS0	Chip Select
5	GPIO11	FSPID	Data in
6	GPIO12	FSPICLK	Clock
7	GPIO13	FSPIQ	Data Output
8	GPIO14	FSPIWP	Write protect

4 RF Characteristics

The module provides 2 RF interfaces, and the impedance of the RF ports is 50Ω.

4.1 Antenna interface

Table 4-1 RF Pin Definitions

Pin number	Pin Definition	Function Description
1	LoRa_RF	LoRa RF antenna interface
36	WIFI&BLE_RF	WiFi & BLE RF Antenna Interface

4.2 Wi-Fi Performance

Table 4-2 Wi-Fi Performance Parameters

Performance	Description	
Wireless Standard	IEEE 802.11b/g/n(@2.4GHz)	
Operating frequency	2.400GHz ~ 2.4835GHz	
Channel	2.4GHz: Ch1 ~ Ch13	
Modulation method	802.11b	DQPSK, DBPSK, CCK
	802.11g/n: OFDM	64-QAM, 16-QAM, QPSK, BPSK
Transmit power	802.11b/1Mbps	19.5 dBm@ EVM ≤ -10.5dB
	802.11b/11Mbps	19.5 dBm@ EVM ≤ -10.5dB
	802.11g/6Mbps	18 dBm@ EVM ≤ -5dB
	802.11g/54Mbps	16.5 dBm@ EVM ≤ -25dB
	802.11n/MCS0(20/40M)	17.5 dBm@ EVM ≤ -5dB
	802.11n/MCS7(20/40M)	16 dBm@ EVM ≤ -27dB
Frequency tolerance	±20ppm	
Receive sensitivity	802.11b/1Mbps	-96 dBm, typical
	802.11b/11Mbps	-86.5 dBm, typical
	802.11g/6Mbps	-90.5 dBm, typical

	802.11g/54Mbps	-74.5 dBm, typical
	802.11n/MCS0(20M)	-90.5 dBm, typical
	802.11n/MCS7(20M)	-72 dBm, typical
	802.11n/MCS0(40M)	-88 dBm, typical
	802.11n/MCS7(40M)	-69.5 dBm, typical

Remarks

- Test conditions for the data in Table 4-2: temperature 25°C, operating voltage 3.3 V.

4.3 BLE Performance

The module supports BLE mode.

Table 4-3 LE Performance Parameters

Performance	Description	
Wireless standard	BLE5.0	
Operating frequency	2.4GHz ~ 2.4835GHz	
Channel	LE: Ch0 ~ Ch39	
Modulation method	GFSK	
Transmit power	-24 ~ 20dBm	
Receive sensitivity	BLE 1M@ PER≤30.8%	-95.5 dBm, typical
	BLE 2M@ PER≤30.8%	-91.5 dBm, typical
Maximum input level	BLE 1M@ PER≤30.8%	8dBm, Typical
	BLE 2M@ PER≤30.8%	3dBm, Typical

Remarks

- Test conditions for the data in Table 4-3: temperature 25°C, operating voltage 3.3 V.

4.4 LoRa Performance

Table 4-4 LoRa Performance Parameters

Performance	Description	
Wireless Standard	Generation 4 LoRa	
Operating frequency	860 ~ 930MHz	
Modulation method	LoRa/GFSK/FLRC	
Transmit Power @ Sub-GHz	Power Setting = 22dBm	21dBm, typical
Receive Sensitivity @SubG	Rx Boosted,BW=125KHz,SF=7,CR 4/5,PER≤10%	-125dBm, typical

Remarks

1. Test conditions for the data in Table 4-4: temperature 25°C , center frequency: 915 MHz, operating voltage 3.3 V;
2. Users should configure and use the device according to the operating frequency bands permitted by the local regulations of the target terminal market. Please be sure to comply with local regulations. If the device is used within frequency bands not permitted by regulations, our company assumes no responsibility. For applications in the domestic terminal market, please refer to the “Catalog and Technical Requirements for Micropower Short-Range Radio Transmission Equipment.”

4.5 Antenna Circuit Reference Design

When users use the module, it is recommended to add a π -type matching circuit between the module's antenna interface and the antenna interface on the user's baseboard. The typical antenna matching circuit and initial parameters are shown in the figure below. R1 can be populated with a 0 Ω resistor or a 100 pF capacitor, while the positions for capacitors C1 and C2 are not populated by default. The recommended component package sizes are 0201 or 0402.

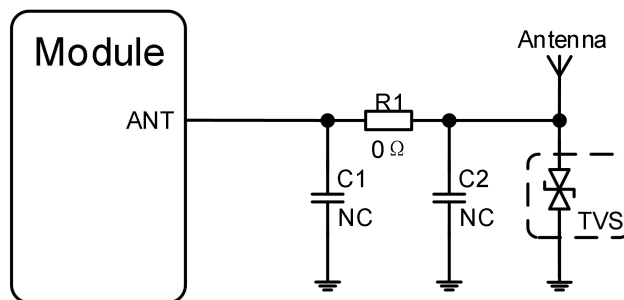


Figure 4.1 RF Interface Reference Circuit

If an external antenna is used, or if the user can touch the antenna, it is recommended to reserve a TVS diode to enhance electrostatic protection. The junction capacitance of this TVS should be as small as possible, preferably below 0.5 pF. Since the parasitic capacitance of the TVS may affect antenna performance, it is recommended to retune the antenna after adding the TVS diode.

The routing requirements between ANT and the user antenna must comply with the 50Ω RF characteristic impedance requirement. At the same time, the RF trace length should be as short as possible to ensure that the insertion loss of the RF trace is minimized. Detailed routing requirements are provided in the next section.

4.6 RF Signal Line Routing Guidelines

◆ RF Trace Design Requirements

The characteristic impedance of the RF signal line in the system where this module is applied should be controlled at 50 Ω. Generally, the impedance of the RF signal line is determined by the dielectric constant ϵ_r of the material, the trace width W , the distance to ground D , and the thickness H of the reference ground plane. In IoT applications, the design of PCB characteristic impedance is usually implemented using a coplanar waveguide structure, which helps provide better shielding for the RF signal line and enables a more compact design with higher integration. The figure below describes the structural requirements for PCB designs with different numbers of layers.

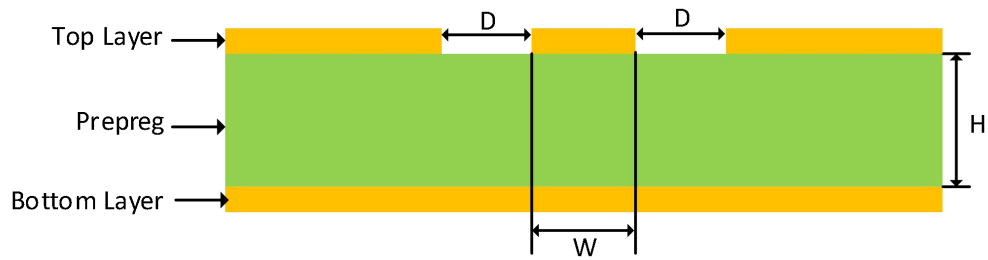


Figure 4.2 Coplanar Waveguide Structure of a Two-Layer PCB Board

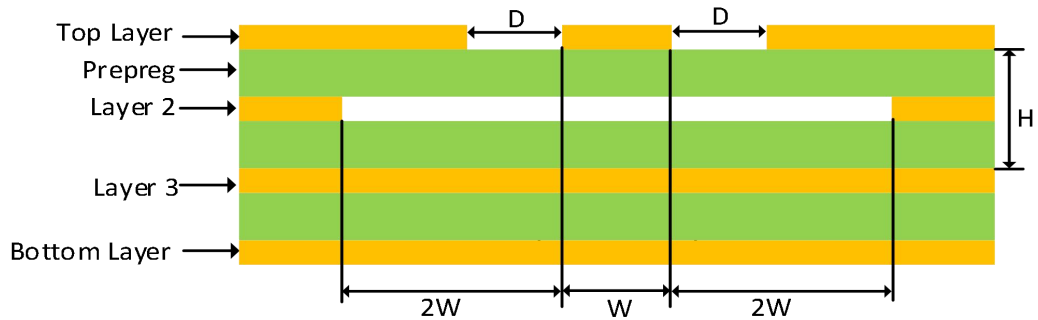


Figure 4.3 Four-layer PCB coplanar waveguide structure (reference ground is the third layer)

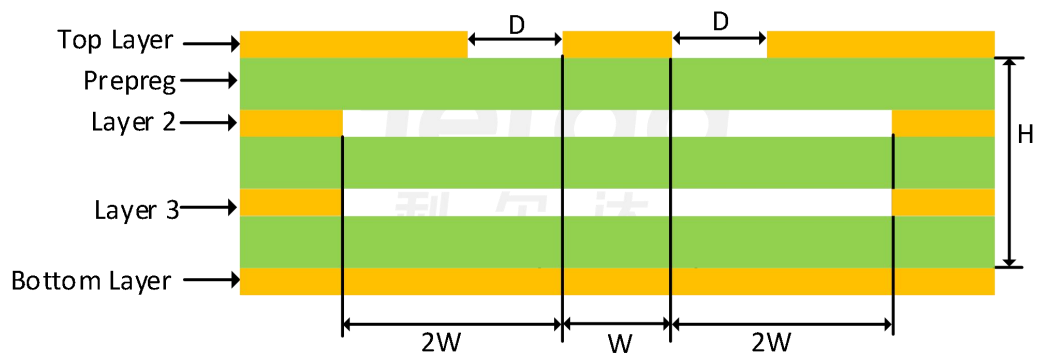


Figure 4.4 Four-layer PCB coplanar waveguide structure (the reference ground is the fourth layer)

The control method for 50Ω impedance in LAYOUT design can use the Polar Si9000 design software tool. The calculation method shown below takes a finished PCB thickness of 1.6 mm as an example, resulting in an RF trace width $W = 0.65$ mm and trace spacing $D = 0.14$ mm.



Figure 4.5 Reference for the calculation method of 50 Ω impedance

The figure below is a layout diagram of the PCB RF circuit. The recommendations are as follows:

- ◆ The RF trace width W and trace spacing D shall be designed based on the above design results.
- ◆ The three reserved external matching components in the π -type circuit are placed closely together. The reserved components can be placed on the same side or on both sides in the layout design (as shown in the figure).
- ◆ Irregular vias must be placed on the GND planes on both sides of the RF trace to ensure that there are vias on the nearest GND planes on both sides of the RF trace (as shown in the green boxed areas in the figure). It is recommended to have a complete GND plane under the entire RF trace area (as shown in the blue area in the figure).

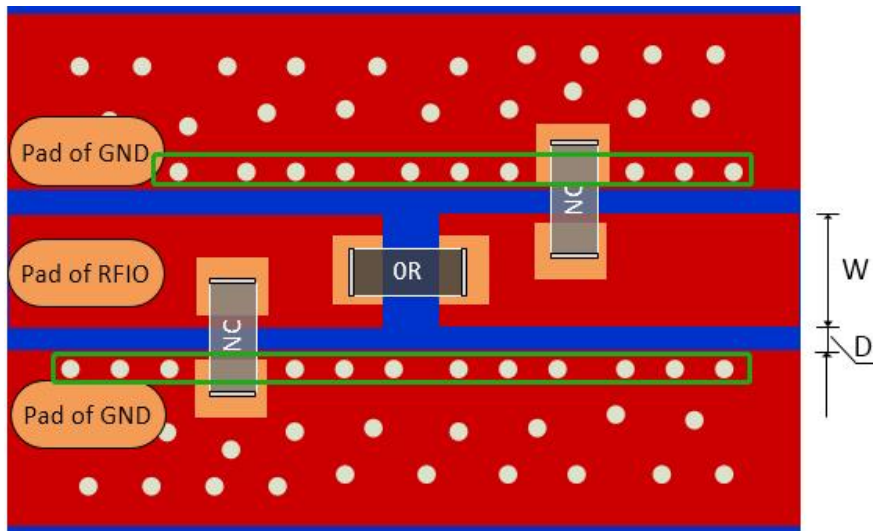


Figure 4.6 Schematic Diagram of RF Trace Layout Design

◆ Routing Design Guidelines for Modules in Products

RF trace layout quality is directly reflected in the module conducted test results, but to ensure that the complete product can achieve maximum performance, antenna design coordination is also required. To optimize antenna performance, follow these PCB design guidelines in PCB design. Guidance and suggestions for complete-product PCBs with different layer counts are provided below:

1) When the product PCB is designed with 2 layers, it is recommended that both the TOP and BOTTOM layers directly beneath the module be GND layers. Traces that need to be routed out from the module should avoid passing directly beneath the module and should be routed out from the outside of the module.

2) When the product PCB is designed with 4 layers, it is recommended that the traces routed out from the module be placed on the third or fourth layer, while the first and second layers should be reserved as complete GND reference layers for the module.

4.7 Antenna Design Requirements

Basic requirements for antennas used in module terminals: Select an antenna that matches the module's operating frequency band. The antenna impedance must be 50Ω . Minimize insertion loss within the operating band. The voltage standing wave ratio (VSWR) should be ≤ 2 . The better the antenna performance, the higher the module efficiency and

the wider the coverage range.

Since the module has two external antenna interfaces, in addition to the basic antenna requirements above, the following additional requirements also apply:

1. Scenario where LoRa and WiFi & BLE work simultaneously:

1) The distance between the dual antennas is recommended to be greater than 10 cm, and greater than 20 cm is recommended if conditions allow. When the distance is relatively short, one antenna may be located in the radiating near field of the other antenna, or even in the reactive near field, affecting the impedance and resonant frequency of both antennas.

2) It is recommended that the isolation between the dual antennas be greater than 30 dB (i.e., the S21 between the two antennas should be less than -30 dB). Although the operating frequency bands of the dual antennas are different, the out-of-band spurious emissions and high-power near field generated when the transmitter at one end is transmitting will also raise the noise floor, affecting the SNR of the receiver at the other end.

3) The antennas need joint tuning. When two antennas are close to each other, due to mutual coupling, when one antenna radiates, its electromagnetic field will also induce current on the other antenna, affecting each other's antenna impedance. Therefore, an antenna that has been tuned individually may experience resonant frequency shift and performance degradation after the other antenna is installed. Therefore, antenna tuning needs to be performed with the dual antennas coexisting, so that the antenna impedance and resonant frequency meet the basic antenna requirements described above.

2. Time-division use of LoRa and WiFi & BLE:

1) Since the two RF systems are used in time division, when the transmitter at one end is transmitting, the receiver at the other end is not operating. Therefore, the isolation requirement for the dual antennas is no longer required, and the distance requirement can be relaxed. However, it is still recommended to keep the distance greater than 10 cm to avoid placing one antenna within the induction near field of the other antenna, which would

greatly affect the impedance and resonant frequency of both antennas.

2) The antenna still needs joint debugging.

4.8 RF Connector Recommendations

Two conventional methods for connecting an antenna:

◆ Pad soldering: One end of the antenna is directly soldered to the product's antenna output port using a high-frequency cable.

◆ RF tuner: Uses IPX and SMA terminal connection methods. For the IPX terminal, Hirose's UF.L-R-SMT connector is recommended. Physical image of the IPX terminal,

physical image of the SMA connector.



4.9 Baseboard layout considerations

Description of the test points and thermal pads on the bottom layer of the module:

There are 2 exposed-copper heat dissipation pads (EPADs) and 7 test points in the BOTTOM layer area of the module, as shown in the figure below.

The EPAD may be left unsoldered to the baseboard, but soldering it to the GND on the baseboard can provide better thermal performance. If you want to solder the EPAD to the baseboard, please ensure that an appropriate amount of solder paste is used to avoid excessive solder paste causing the distance between the module and the baseboard to be too large, which may affect the contact between the pins and the baseboard.

The positions on the baseboard corresponding to the 7 test points must not have vias or exposed copper. It is recommended that the routing or exposed copper on the baseboard avoid the red boxed areas shown in the figure below. If routing is required, it is recommended to add solder mask blocks at the corresponding positions.

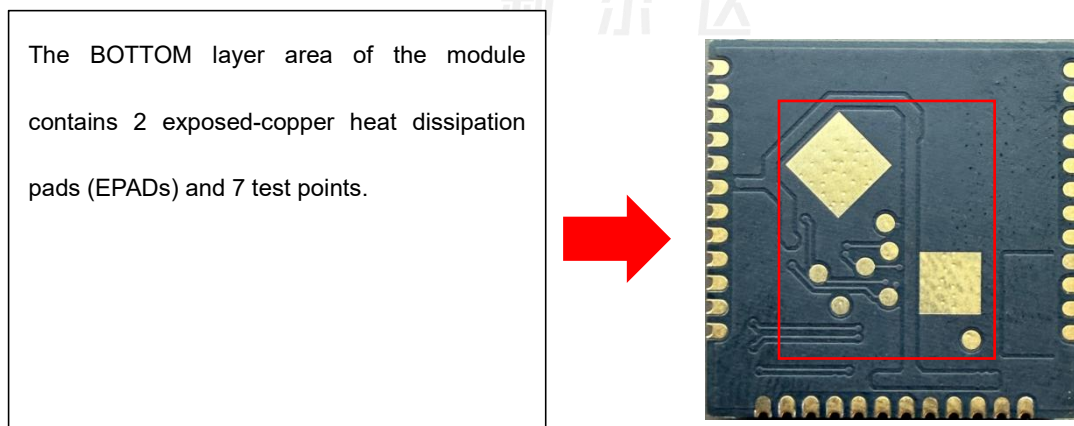


Figure 4.7 Positions of Test Points and Pads on the Bottom of the Module

Baseboard layout requirements:

There are no high-speed signals or sensitive signal traces on the BOTTOM layer of the

module, but it is still recommended that the routing design on the TOP layer of the baseboard avoid the module to prevent unexpected influencing factors.

There are no excessive cutout requirements in the baseboard design. Apart from the general requirement mentioned earlier to avoid interference sources, the baseboard can almost be fully copper-poured.

5 Electrical performance and reliability

5.1 Absolute Maximum Ratings

Table 5-1 Product Limit Parameters

Main Parameters	Performance		Remarks
	Minimum value	Maximum value	
Power Supply Voltage (V)	-0.3	3.6	-
IO Voltage (V)	-0.3	VCC	-
Operating Temperature 1 (°C)	-40	65	-
V _{ESD}	±2kV		VCC&ANT ESD performance

Note

1. The operating ambient temperature of the module' s main chip ESP32-S3R8 is -40°C to +65°C. If the PSRAM ECC function is enabled, the module' s maximum ambient temperature can be increased to 85°C, but the available PSRAM capacity will be reduced by 1/16.

5.2 DC Electrical Characteristics

Table 5-2 DC Electrical Characteristics (3.3 V, 25°C)

Parameter	Description	Minimum value	Typical value	Maximum value	Unit
CIN	Pin capacitance	-	2	-	pF
V _{IH}	High-level input voltage	0.75*VCC	-	VCC+0.3V	V
V _{IL}	Low-level input voltage	-0.3	-	0.25*VCC	V
I _{IH}	High-level input current	-	-	50	nA
I _{IL}	Low-level input current	-	-	50	nA
V _{OH}	High-level output voltage	0.8*VCC	-	-	V
V _{OL}	Low-level output voltage	-	-	0.1*VCC	V

I_{OH}	High-level pull-up current (VCC = 3.3V, VOH >= 2.64V, PAD_DRIVER=3)	-	40	-	mA
I_{OL}	Low-level sink current (VCC = 3.3 V, VOL = 0.495 V, PAD_DRIVER=3)	-	28	-	mA
R_{PU}	Internal weak pull-up resistor	-	45	-	k Ω
R_{PD}	Internal weak pull-down resistor	-	45	-	k Ω
V_{IH_nRST}	Chip reset release voltage (the EN pin should meet the voltage range)	0.75*VCC	-	VCC+0.3V	V
V_{IL_nRST}	Chip reset voltage (the EN pin should meet the voltage range)	-0.3	-	0.25*VCC	V

5.3 Power Consumption Characteristics

Table 5-3 Module Power Consumption Parameters

Main parameters	Performance			Remarks
	Minimum value	Typical value	Maximum value	
Transmit Current (mA)	/	223.12	/	11b 1M @50% duty cycle
	/	225.69	/	11b 11M @50% duty cycle
	/	217.15	/	11g 6M @50% duty cycle
	/	197.19	/	11g 54M @50% duty cycle
	/	206.15	/	11n20M MCS0 @50% duty cycle
	/	192.41	/	11n20M MCS7 @50% duty cycle
	/	179.91	/	11n40M MCS0 @50% duty cycle
	/	169.86	/	11n40M MCS7 @50% duty cycle
	/	357.50	/	BLE 1M @ 20dBm
	/	205.54	/	BLE 1M @ 10dBm
	/	168.35	/	BLE 1M @ 0dBm
	/	267.57	/	BLE 2M @ 20dBm
	/	159.05	/	BLE 2M @ 10dBm
	/	136.60	/	BLE 2M @ 0dBm
	/	155	/	LoRa sub-G@22dBm

Receive Current (mA)	/	100.11	/	802.11b
	/	100.51	/	802.11g
	/	100.75	/	802.11n,HT20
	/	103.79	/	802.11n,HT40
	/	104.07	/	BLE 1M
	/	104.40	/	BLE 2M
	/	51	/	LoRa@868MHz
Sleep current (μA)	/	9.5	/	Deep-sleep @RTC memory powered on, RTC peripherals powered off

Remarks

- Test conditions for the data in Table 5-3: temperature 25°C, operating voltage 3.3 V.

5.4 Electrostatic Protection

ESD Protection Instructions:

- In module applications, static electricity generated by sources such as human electrostatic discharge and charged friction between microelectronics may discharge to the module through various paths, potentially causing certain damage to the module. Therefore, ESD protection should be taken seriously.
- During R&D, production assembly, testing, and other processes, especially in product design, ESD protection measures should be implemented. For example, anti-static protection should be added at circuit design interfaces and points that are susceptible to damage or impact from electrostatic discharge; test equipment must be properly grounded; anti-static gloves should be worn during production, etc.
- To minimize the impact of static electricity and surges as much as possible, antennas and coaxial cables connected to test instruments must be plugged in or unplugged when the module is powered off.

6 Mechanical Dimensions

6.1 Mechanical dimensions

Overall dimensions	20 mm × 20 mm × 2.5 mm
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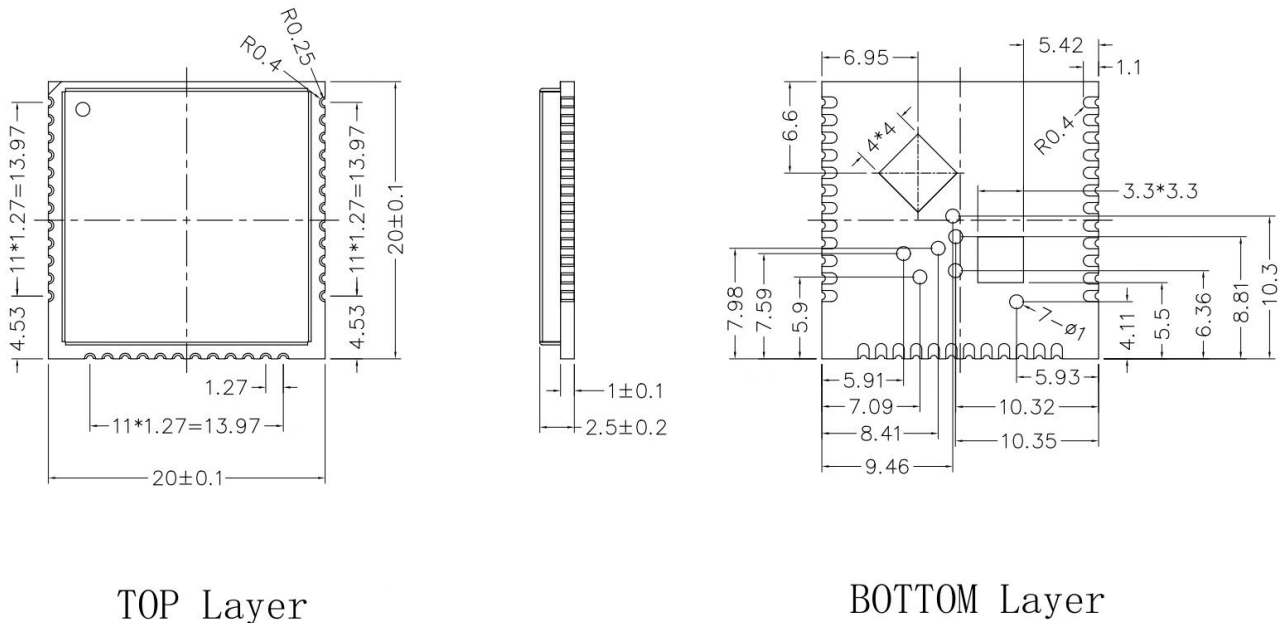


Figure 6.1 Module Dimensions Diagram

Remarks

There is a cutting tolerance in the longitudinal rib connection of the module, and the module length range is [19.7 mm - 20.3 mm].

7 Production and Packaging Information

7.1 Production welding

7.1.1 Production Guide

It is recommended to mount the stamp-hole package module using an SMT machine and complete the mounting within 24 hours after unpacking. Otherwise, it should be vacuum-packed again to prevent moisture absorption that may cause poor mounting.

If the package contains a humidity indicator card, it is recommended to determine whether the module needs to be baked based on the indication of the humidity card. The baking conditions are as follows:

Baking temperature: $125^{\circ}\text{C} \pm 5^{\circ}\text{C}$;

The alarm temperature is set to 130°C .

After cooling to below 36°C under natural conditions, SMT mounting can be carried out.

If the unpacking time exceeds 3 months, special attention should be paid to whether the product has been affected by moisture. Because the PCB uses an immersion gold process, more than 3 months may lead to pad oxidation, which may cause issues such as cold solder joints and missing solder during SMT assembly.

To ensure the pass rate of reflow soldering, it is recommended to randomly select 10% of the products for visual inspection and AOI inspection during the first SMT process, to ensure the rationality of oven temperature control, component pickup method, and placement method.

Operators at all workstations must wear anti-static gloves throughout the entire production process.

7.1.2 Module position requirements on the baseboard

It is recommended that the solder mask thickness at the module position on the

baseboard be less than 0.02 mm to avoid excessive thickness, which may raise the module and prevent effective contact with the solder paste, affecting soldering quality. In addition, note that no other components should be placed within 2 mm around the module position on the interface board to ensure module repairability.

7.1.3 Stencil Aperture Design

The principle for selecting the thickness of the stencil on the baseboard is based on comprehensive consideration of the package types of the components on the board. The following requirements should be given special attention:

The module pad area can be locally thickened to 0.15–0.20 mm to avoid cold solder joints.

7.1.4 Production Precautions

- During the production process, all operators must wear anti-static gloves.
- Do not exceed the specified baking time during baking.
- It is strictly forbidden to add explosive, flammable, or corrosive substances during baking.
- During baking, the modules should be placed in a high-temperature tray, with air circulation maintained between the modules.
- During baking, the oven door must be properly closed to ensure the oven is sealed and prevent heat from escaping.
- Try not to open the oven door while the oven is running. If it must be opened, minimize the time the door remains open.
- After baking is complete, wait until the module naturally cools to below 36°C before wearing anti-static gloves to remove it, to avoid burns.
- During operation, strictly prevent the bottom surface of the module from coming into contact with water or contaminants.

7.1.5 Reflow Soldering Work Instructions

This work instruction is only applicable to lead-free operations and is for reference only.

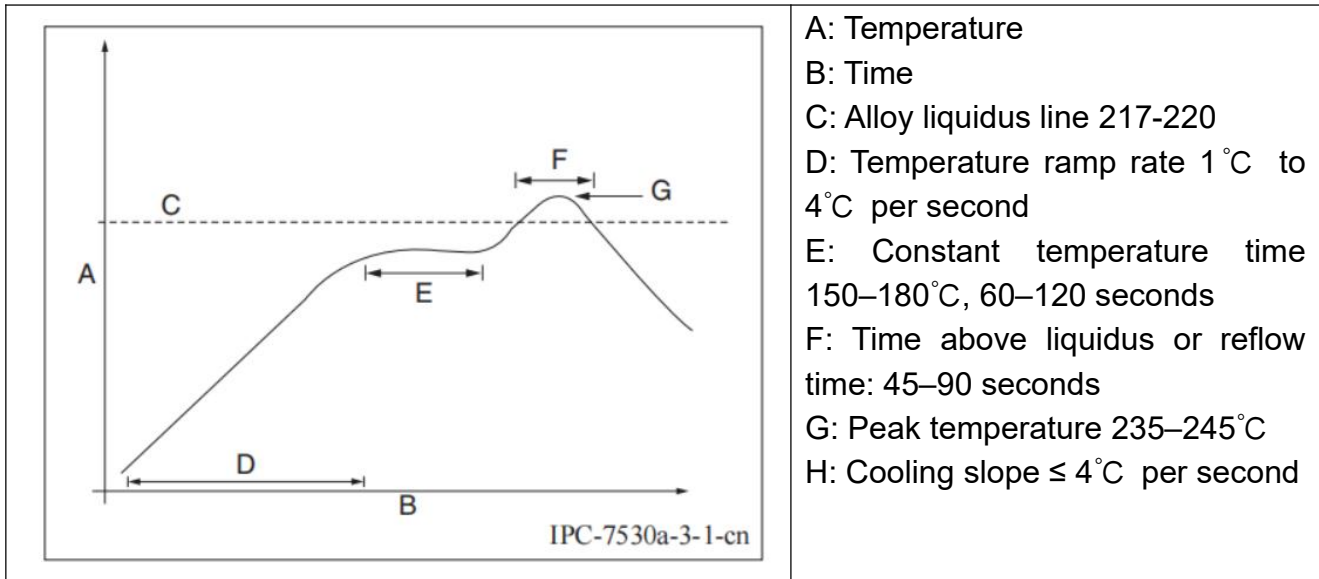


Figure 7.1 Reflow Soldering Work Instruction

Remarks

◆ To ensure module soldering quality, the recommended stencil thickness corresponding to the module pad area is 0.18 - 0.2 mm.

◆ To prevent the module from being damaged due to repeated heating, it is strongly recommended that customers mount the module during second-side production or pass it through reflow soldering only once.

◆ During production, welding, and other processes involving contact with the module, do not use any organic solvents (such as alcohol, isopropyl alcohol, trichloroethylene, etc.) to wipe the shielding cover; otherwise, the shielding cover may rust or the QR code may fail to be read properly.

◆ Do not ultrasonically clean the module under any circumstances, as this may damage the chips inside the module.

7.2 Packaging Specifications

7.2.1 Packaging method

Model	Packaging method	Full case (PCS)	Minimum Package Quantity (PCS)	Number of reels per box
L-LRMAM36-FANN4	Tape and reel	1500	300	5

7.2.2 Tape dimensions

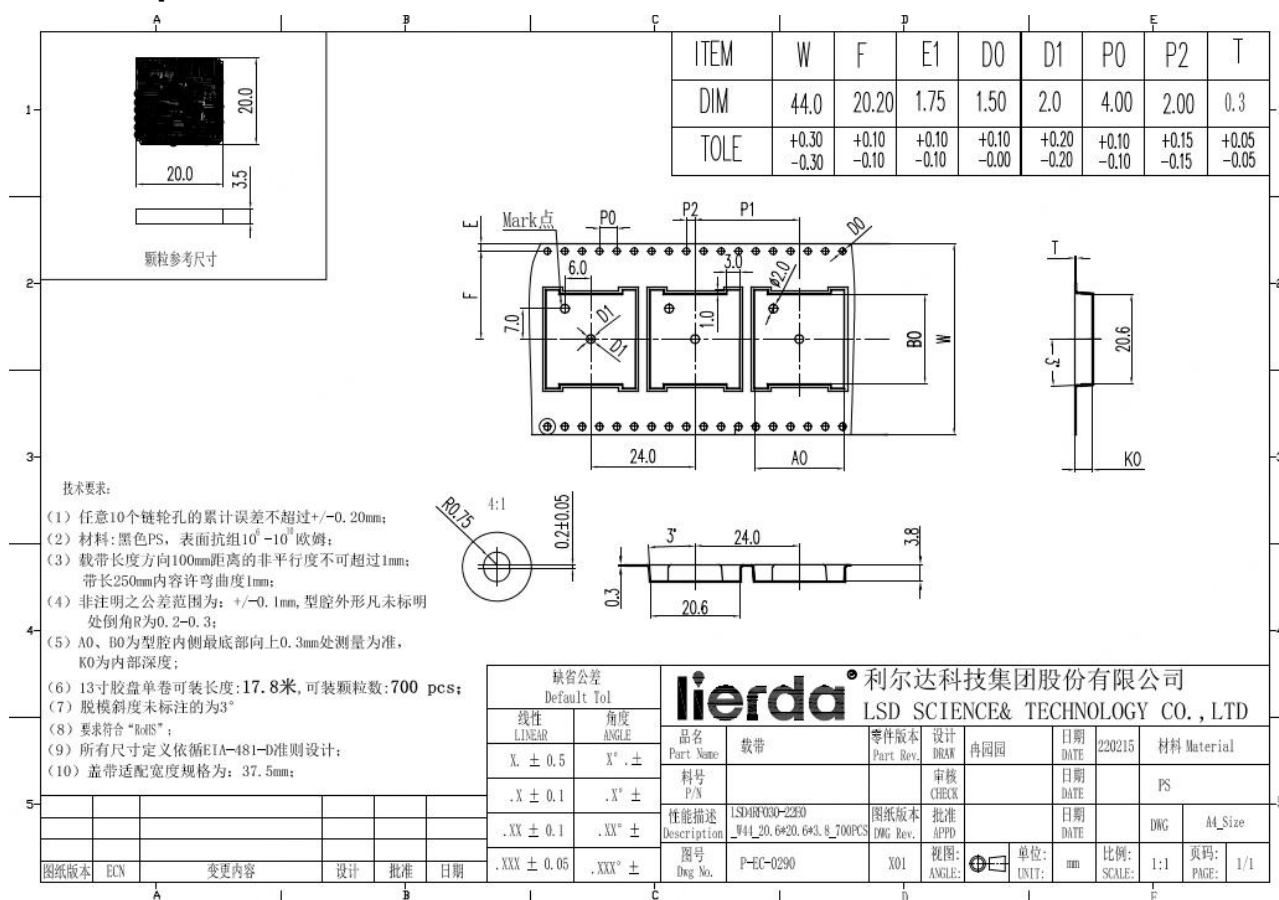


Figure 7.2 Carrier Tape Dimensions